

AFRILANG Stdlib

std/c/gameastar

Français. Bibliothèque AFRILANG 'std/c/gameastar' (niveau complex, catégorie complex). Elle expose 7 fonction(s) : heuristic, astar, pathCost, reconstruct, manhattan, octile, openSetSize.

```
'import "std/c/gameastar"' · 'use gameastar'
```

```
- 'heuristic(ax number, ay number, bx number, by number) → number'  
- 'astar(grid list number, width number, start number, goal number) →  
list number' - 'pathCost(grid list number, width number, start num-  
ber, goal number) → number' - 'reconstruct(cameFrom list number,  
current number) → list number' - 'manhattan(ax number, ay number,  
bx number, by number) → number' - 'octile
```

English. AFRILANG library 'std/c/gameastar' (tier complex, category complex). Exports 7 function(s): heuristic, astar, pathCost, reconstruct, manhattan, octile, openSetSize.

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'import "std/c/gameastar"' · 'use gameastar'
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goal number) → number' - 'reconstruct(cameFrom list number, current  
number) → list number' - 'manhattan(ax number, ay number, bx number,  
by number) → number' - 'octile(ax number, a
```

Catégorie / Category	Modules complex (c/)
Niveau / Tier	complex
Fonctions / Functions	7

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Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/gameastar' (niveau complexe, catégorie complexe). Elle expose 7 fonction(s) : heuristic, astar, pathCost, reconstruct, manhattan, octile, openSetSize.

'import "std/c/gameastar"' · 'use gameastar'

```
- `heuristic(ax number, ay number, bx number, by number) → number`
- `astar(grid list number, width number, start number, goal number) → list`
- `pathCost(grid list number, width number, start number, goal number) → number`
- `reconstruct(cameFrom list number, current number) → list`
- `manhattan(ax number, ay number, bx number, by number) → number`
- `octile(ax number, ay number, bx number, by number) → number`
- `openSetSize(grid list number) → number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/gameastar"
use gameastar
```

Niveau : complexe · Catégorie : Modules complex (c/)
Domaines avancés – graphes, NLP, réseaux complexes.

3. Référence API

- heuristic(ax number, ay number, bx number, by number) → number•
- astar(grid list number, width number, start number, goal number) → list•
- pathCost(grid list number, width number, start number, goal number) → number•
- reconstruct(cameFrom list number, current number) → list•
- manhattan(ax number, ay number, bx number, by number) → number•
- octile(ax number, ay number, bx number, by number) → number•
- openSetSize(grid list number) → number

4. Exemples d'utilisation

```
import "std/c/gameastar"
use gameastar
```

```
create result = heuristic(/* args */)
say result
```

```
create result = astar(/* args */)
say result
```

```
create result = pathCost(/* args */)
say result
```

```
create result = reconstruct(/* args */)
say result
```

```
create result = manhattan(/* args */)
say result
```

```
create result = octile(/* args */)
say result
```

5. Bonnes pratiques

- Préférez ``import "std/c/gameastar"`` en tête de fichier.
- Lancez ``afri-lang check`` avant de livrer.
- Combinez avec les modules `stdlib` de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

6. Annexe — source

module gameastar

```
function heuristic(ax number, ay number, bx number, by number) returns number
end
```

```
function astar(grid list number, width number, start number, goal number) returns
  list number
end
```

```
function pathCost(grid list number, width number, start number, goal number) returns
  number
end
```

```
function reconstruct(cameFrom list number, current number) returns list number
end
```

```
function manhattan(ax number, ay number, bx number, by number) returns number
end
```

```
function octile(ax number, ay number, bx number, by number) returns number
end
```

```
function openSetSize(grid list number) returns number
end
end
```

English

1. Overview

AFRILANG library 'std/c/gameastar' (tier complex, category complex). Exports 7 function(s): heuristic, astar, pathCost, reconstruct, manhattan, octile, openSetSize.

'import "std/c/gameastar"' · 'use gameastar'

```
- `heuristic(ax number, ay number, bx number, by number) → number`
- `astar(grid list number, width number, start number, goal number) → list number`
- `pathCost(grid list number, width number, start number, goal number) → number`
- `reconstruct(cameFrom list number, current number) → list number`
- `manhattan(ax number, ay number, bx number, by number) → number`
- `octile(ax number, ay number, bx number, by number) → number`
- `openSetSize(grid list number) → number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/gameastar"
use gameastar
```

Tier: complex · Category: Complex modules (c/)
Advanced domains – graphs, NLP, complex networks.

3. API reference

- heuristic(ax number, ay number, bx number, by number) → number•
- astar(grid list number, width number, start number, goal number) → list•
- pathCost(grid list number, width number, start number, goal number) → number•
- reconstruct(cameFrom list number, current number) → list•
- manhattan(ax number, ay number, bx number, by number) → number•
- octile(ax number, ay number, bx number, by number) → number•
- openSetSize(grid list number) → number

4. Usage examples

```
import "std/c/gameastar"
use gameastar
```

```
create result = heuristic(/* args */)
say result
```

```
create result = astar(/* args */)
say result
```

```
create result = pathCost(/* args */)
say result
```

```
create result = reconstruct(/* args */)
say result
```

```
create result = manhattan(/* args */)
say result
```

```
create result = octile(/* args */)
say result
```

5. Best practices

- Prefer ``import "std/c/gamestar"`` at the top of your file.
- Run ``afrilang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

6. Appendix — source

module gamestar

```
function heuristic(ax number, ay number, bx number, by number) returns number
end
```

```
function astar(grid list number, width number, start number, goal number) returns
  list number
end
```

```
function pathCost(grid list number, width number, start number, goal number) returns
  number
end
```

```
function reconstruct(cameFrom list number, current number) returns list number
end
```

```
function manhattan(ax number, ay number, bx number, by number) returns number
end
```

```
function octile(ax number, ay number, bx number, by number) returns number
end
```

```
function openSetSize(grid list number) returns number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/gamestar"`
- Vérification : `afrilang check`
- Documentation : <https://afrilang.dev/stdlib/std/c/gamestar/>

Source (extrait)

```
module gameastar

function heuristic(ax number, ay number, bx number, by number) returns number
end

function astar(grid list number, width number, start number, goal number) returns
    list number
end

function pathCost(grid list number, width number, start number, goal number) returns
    number
end

function reconstruct(cameFrom list number, current number) returns list number
end

function manhattan(ax number, ay number, bx number, by number) returns number
end

function octile(ax number, ay number, bx number, by number) returns number
end

function openSetSize(grid list number) returns number
end
end
```